

Formula Sheet Axioms of Probability (2)

① If $E_1, E_2, \dots$ are the events then union of all of their events are denoted by $\bigcup_{n=1}^{\infty} E_n$

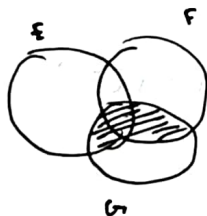
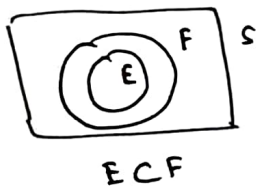
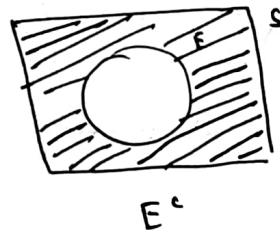
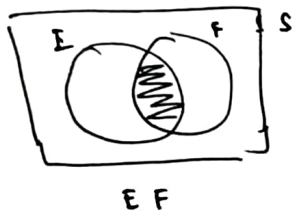
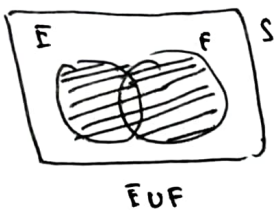
② Intersection of Events E_n is denoted by $\bigcap_{n=1}^{\infty} E_n$

③ Laws of Sets

Commutative $E \cup F = F \cup E$; $E \cap F = F \cap E$

Associative $(E \cup F) \cup G = E \cup (F \cup G)$; $(E \cap F) \cap G = E \cap (F \cap G)$

Distributive $(E \cup F) \cap G = EG \cup FG$; $E \cap (F \cup G) = (E \cap F) \cup (E \cap G)$



④ De Morgan's Law

$$(\bigcup_{i=1}^n E_i)^c = \bigcap_{i=1}^n E_i^c$$

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For 2 events E & F

$$(E \cup F)^c = E^c \cap F^c \quad \& \quad (E \cap F)^c = E^c \cup F^c$$

$$(E^c)^c = E$$

$$(5) \quad P(E) = \lim_{n \rightarrow \infty} \frac{n(E)}{n}$$

$n(E)$ = nos of times in the first n repetition that event E occurs

$P(E)$ = Prob. of event E .

(6) 3 Axiom.

Axiom (1) $0 \leq P(E) \leq 1$

Axiom (2) $P(S) = 1$

Axiom (3) For any sequence of mutually exclusive events

$E_1, E_2, \dots$ (i.e. events for which $E_i E_j = \phi$ when $i \neq j$)

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

$P(E)$ = Prob. of events E

(7) Some Simple Propositions

$$1 = P(S) = P(E \cup E^c) = P(E) + P(E^c)$$

Prop (1) $P(E^c) = 1 - P(E)$

Prop (2) If Event E is contained in event F then the Probability of E not greater than F

$$E \subset F \quad P(E) \leq P(F)$$

Prop (3) $P(E \cup F) = P(E) + P(F) - P(E \cap F)$

Prop (4) Probability of "u" of n events taken 1 at a time - the sum of Probability of events taken 2 at a time & the

Sum of Probability taken 3 at a time so on

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = \sum_{i=1}^n P(E_i) - \left(\sum_{i_1 < i_2} P(E_{i_1} \cap E_{i_2}) \right) + \dots + (-1)^{r+1} \sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r}) + \dots + (-1)^{n+1} P(E_1 \cap E_2 \cap \dots \cap E_n)$$

⑧ The General way of writing inc./exc. Principle is

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n (-1)^{r+1} \sum_{i_1 < \dots < i_r} P(E_{i_1} \cap \dots \cap E_{i_r})$$

⑨ Boole's Inequality $P\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n P(E_i)$

⑩ for Any event (E) $P(E) = \frac{\text{nos of outcomes in E}}{\text{" " in S}}$

⑪ Probability on Count. Set of fn

An Sequence of events $\{E_n, n \geq 1\}$ is said to be increasing sequence if.

$$E_1 \subset E_2 \subset \dots \subset E_n \subset E_{n+1} \subset \dots$$

Decreasing Sequence.

$$E_1 \supset E_2 \supset \dots \supset E_n \supset E_{n+1} \supset \dots$$

If $\{E_n, n \geq 1\}$ is an inc. sequence of events, we define new events

denoted by $\lim_{n \rightarrow \infty} E_n$

$$\boxed{\lim_{n \rightarrow \infty} E_n = \bigcup_{i=1}^{\infty} E_n}$$

if $\{E_n, n \geq 1\}$ is an dec. sequence of events we define

$$\lim_{n \rightarrow \infty} E_n$$

$$\boxed{\lim_{n \rightarrow \infty} E_n = \bigcap_{i=1}^{\infty} E_i}$$

Prop 6 if $\{E_n, n \geq 1\}$ is either an inc. order sequence of events

$$\boxed{\lim_{n \rightarrow \infty} P(E_n) = P\left(\lim_{n \rightarrow \infty} E_n\right)}$$